



दक्षिण बिहार केन्द्रीय विश्वविद्यालय CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination May 2026

Name of Department : Computer Science Programme : M.Sc. (AI)
Session : 2025-2027 Semester : II
Course Title : Software Engineering Course Code : CAI82DE01404
Course Credit : 4 Max. Marks : 70 Duration : 3 hours
Enrolment No. (to be filled by Student) : CUSB2502332006

Instruction: The use of scientific calculator is permitted in this examination.

Section – A

Answer All the questions in not more than 50 words. Each question carries 1 Points.

(1 x 10 = 10 Points)

1. Define generic software with example.
2. What do you understand by the term life cycle model of software development?
3. Explain the need of an SRS.
4. What is meant by software prototyping?
5. What is the difference between module coupling and module cohesion?
6. Distinguish between verification and validation.
7. Discuss the need of Risk Management in software engineering.
8. List two advantages and disadvantages of LOC.
9. What are the objectives of testing?
10. Discuss some of the problems that are faced during maintenance of software.

Section – B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.

(6 x 5 = 30 Points)

1. What is Software Requirement Specification (SRS)? Explain the characteristics of a good Software Requirements Specification.
[01+05=06]
2. a) You are asked to develop an accounting system for CUSB Gaya that replaces the existing system. Suggest the most appropriate software process development model that might be used as a basis for managing system development. Give proper justification.

- b) Under what circumstances is it beneficial to construct a prototype? Does construction of a prototype always increase the overall cost of software development?

[03+03=06]

3. Explain the types of cohesion and coupling in the context of software design.
4. Explain why the best programmers do not always make the best software managers. You may find it helpful to base your answer on the software project management activities.
5. Why is accurate estimation of the effort required for completing a project difficult? Briefly explain the different effort estimations methods that are available.

[01+05=06]

6.

- a) Annual Change Traffic (ACT) for a software system is 15% per year. The development effort is 600 PMs. Compute estimate for Annual Maintenance Effort (AME). If lifetime of the project is 10 years, what is the total effort of the project (using Boehm Model)?
- b) Suppose that a project was estimated to be 500 KLOC. Calculate the effort and development time for each of the three modes of Basic COCOMO model i.e., organic, semidetached and embedded.

[03+03=06]

Section – C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.

(15 x 2 = 30 Points)

1. What is black box testing? Explain the different types of black box testing strategies with example.

[02+13=15]

2.

- (a) Consider a software project with 5 tasks T1-T5. Duration of the 5 tasks (in days) are 15,10,12,25 and 10 respectively. T2 and T4 can start when T1 is complete. T3 can start when T2 is complete. T5 can start when both T3 and T4 are complete. The project is to be completed in the shortest time
- (i) Calculate the early time and late time for each event.
- (ii) Find the critical paths and the critical activities.

[03+03=06]

- (b) What is the purpose of data flow diagrams? Explain by constructing a context flow diagram, level -1 DFD and level -2 DFD for a library management system?

[01+08=09]

3. Write short notes on any two of the following:

- (i) Different types of maintenance
- (ii) Path Testing
- (iii) Use Case Diagram